

Mental Health and Psychosocial Outcomes Among Individuals With Mayer-Rokitansky-Küster-Hauser (MRKH) Syndrome: A Scoping Review

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Abstract

Background: Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome is a rare congenital condition (1 in 4,500–5,000 female births) characterized by uterovaginal agenesis in individuals with a 46,XX karyotype and functional ovaries. Diagnosed during adolescence, MRKH is associated with significant psychosocial burden. Mental health outcomes and coping mechanisms remain insufficiently synthesized, limiting integration into clinical care and public health planning.

Methods: Following Arksey and O'Malley's framework and PRISMA-ScR guidelines, we searched PubMed, Scopus, and EBSCOhost databases (January 2019–March 2026). Eligible studies included primary quantitative, qualitative, and mixed-methods research on psychological outcomes and/or coping mechanisms in MRKH populations.

Results: 34 studies met inclusion criteria (provisional). Psychosexual and relational challenges were most frequently reported (n = 24), followed by reduced QoL, body image and self-esteem (n = 18), broader psychological distress (n = 18), and depression and anxiety (n = 10). Coping mechanisms were documented in 14 studies; healthcare gaps in 9. *Results updated from the submitted abstract after final screening.*

Conclusions: MRKH-related psychosocial burden is substantial yet under-integrated into care models. Multidisciplinary, mental health-inclusive care and greater research investment are urgently needed.

Keywords: MRKH syndrome; mental health; coping; psychosocial outcomes; quality of life; reproductive health

Introduction and Research Question

MRKH syndrome is a rare Müllerian aplasia resulting in congenital absence of the uterus and upper two-thirds of the vagina in chromosomally female individuals (46,XX) with functional ovaries. It affects approximately 1 in 4,500–5,000 female births^{1,2} and is typically diagnosed during adolescence, often during evaluation for primary amenorrhea¹—a critical period for identity formation and social development.

The diagnosis carries profound implications for fertility, psychosexual development, and self-image. Professional guidance identifies psychosocial counseling as a key component of care,¹ yet psychosocial outcomes remain underexplored and insufficiently integrated into clinical and public health frameworks.

Research Question

What mental health outcomes and coping mechanisms are reported among individuals with MRKH syndrome, and what healthcare system gaps exist?

Learning Objectives

- Identify key mental health outcomes—including depression, anxiety, reduced quality of life, and psychosexual challenges—reported among individuals with MRKH syndrome.
- Describe coping mechanisms and psychosocial responses, including adaptive and maladaptive strategies, documented in MRKH populations.

Methods

Category	Description
Study Design	Scoping review following Arksey & O'Malley's five-stage framework ³ ; reported per PRISMA-ScR ⁴
Databases	PubMed; Scopus; EBSCOhost (MEDLINE, CINAHL, APA PsycInfo, Women's Studies International)
Years Included	January 2019 – March 2026 (earlier records removed at screening); searches run March 29, 2026
Inclusion Criteria	Primary quantitative, qualitative, and mixed-methods studies reporting psychological outcomes and/or coping mechanisms in MRKH
Exclusion Criteria	Studies focused solely on anatomical, surgical, or fertility outcomes
Screening	516 identified; 192 duplicates removed; 324 screened; 278 excluded; 46 to full text (12 pending); 34 included (provisional). AI-assisted screening; human verification in progress
Synthesis	Abstract-level data charting; frequency counts by outcome domain and thematic grouping of coping mechanisms and healthcare gaps

MRKH Syndrome at a Glance

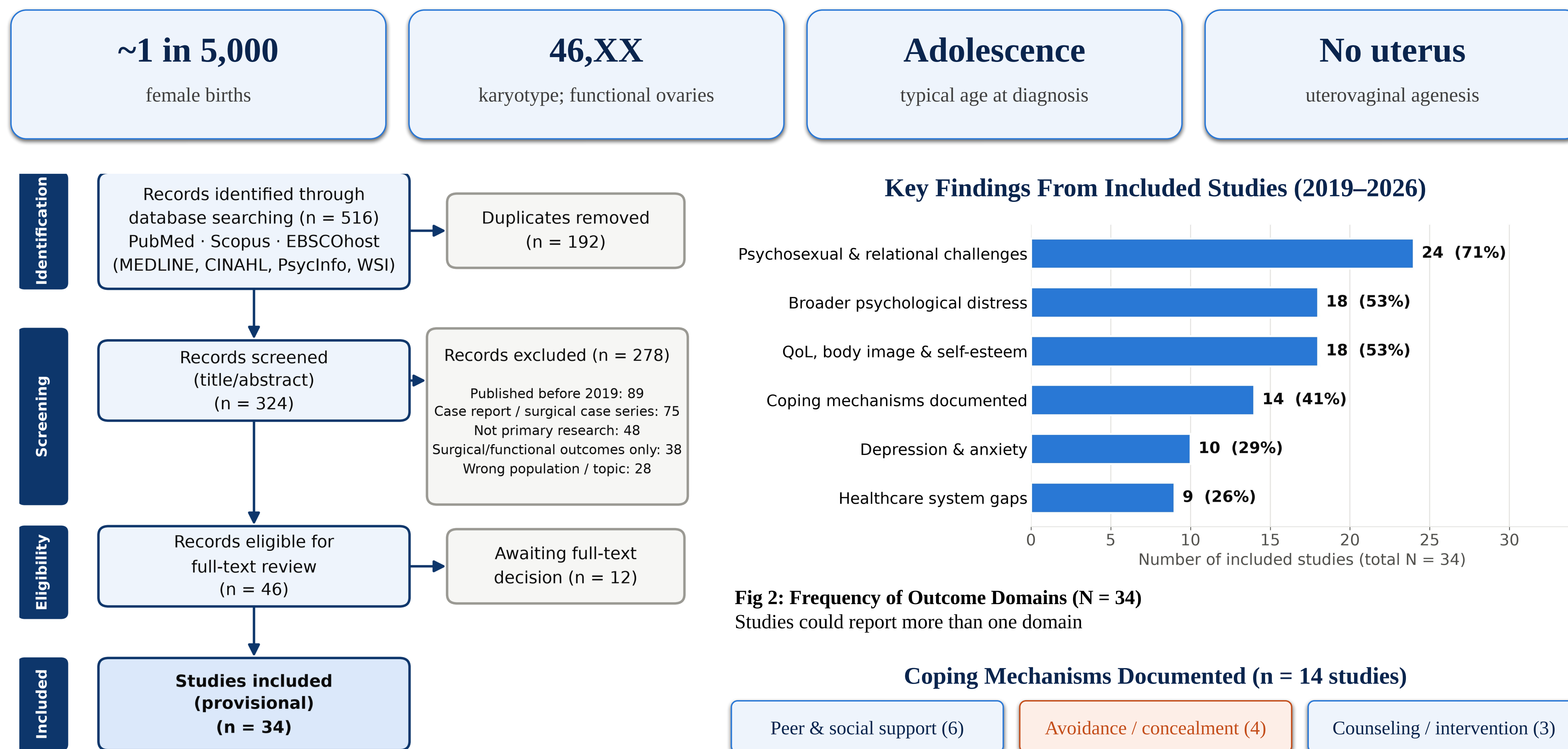


Fig 1: PRISMA-ScR Flow Diagram
(Adapted from Tricco et al., 2018)

Key Findings From Included Studies (2019–2026)

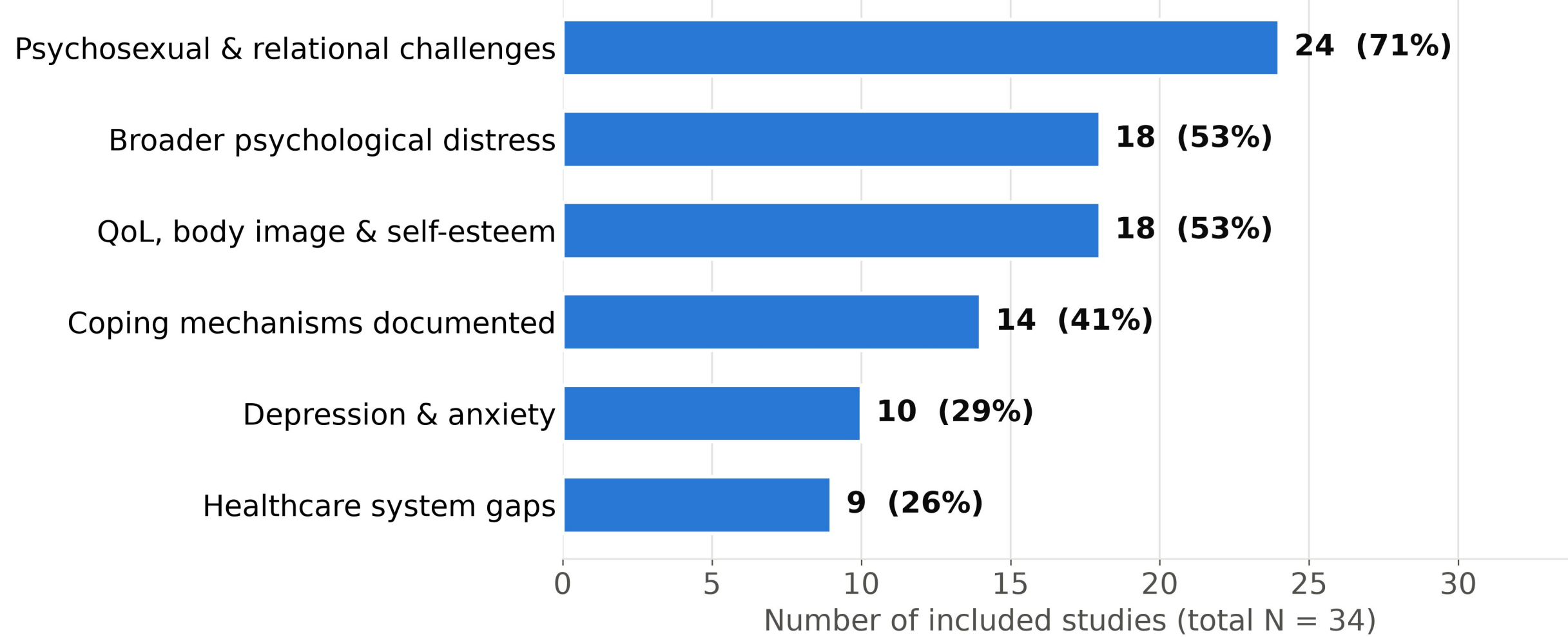


Fig 2: Frequency of Outcome Domains (N = 34)
Studies could report more than one domain

Coping Mechanisms Documented (n = 14 studies)

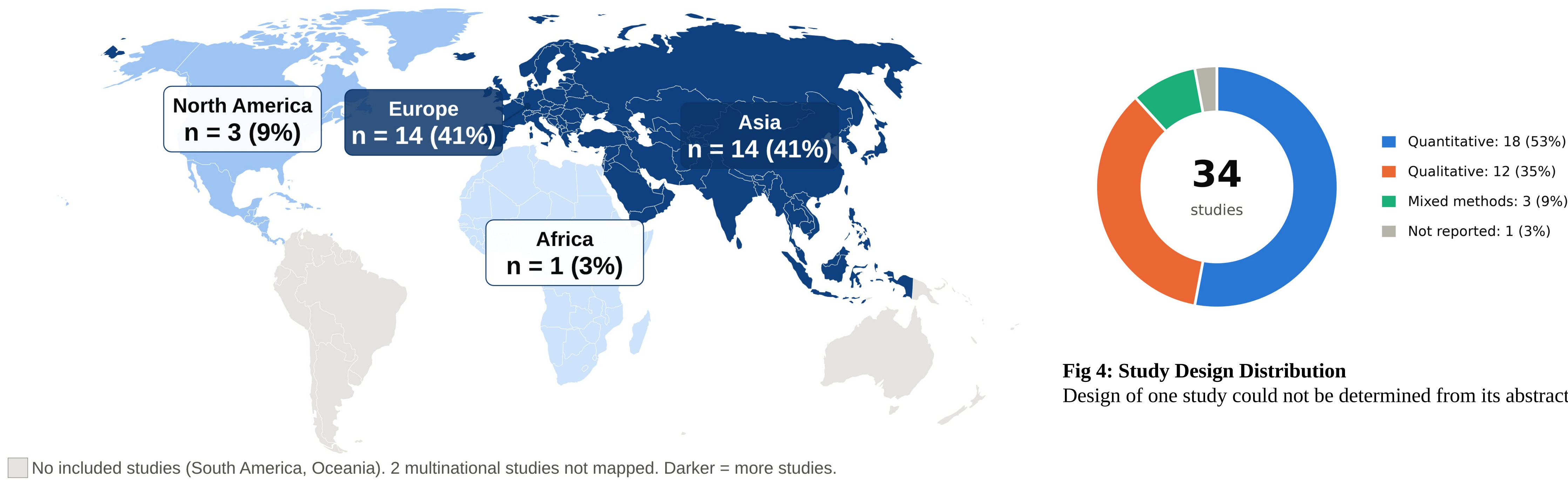
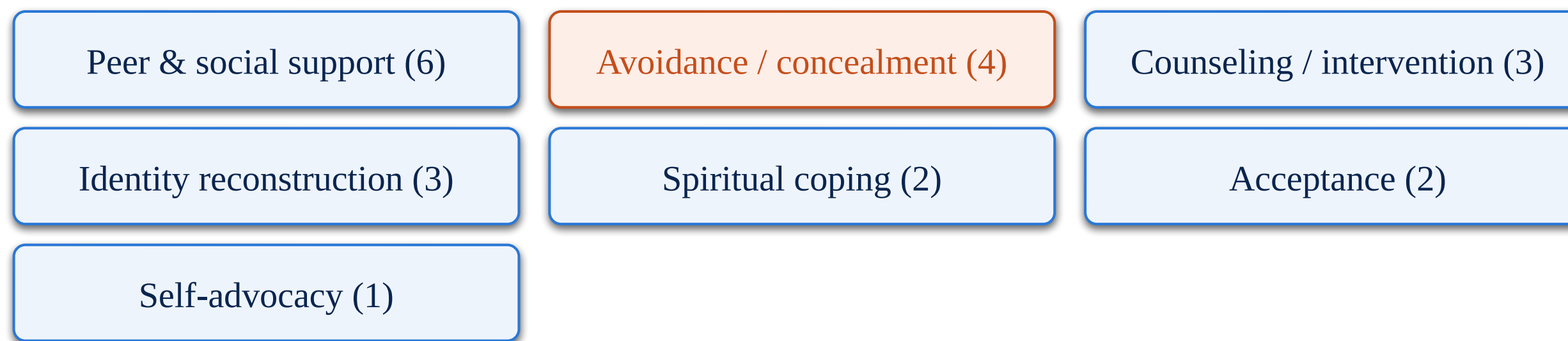


Fig 3: Geographic Distribution of Included Studies
Europe 14, Asia 14, North America 3, Africa 1, multinational 2; none from South America

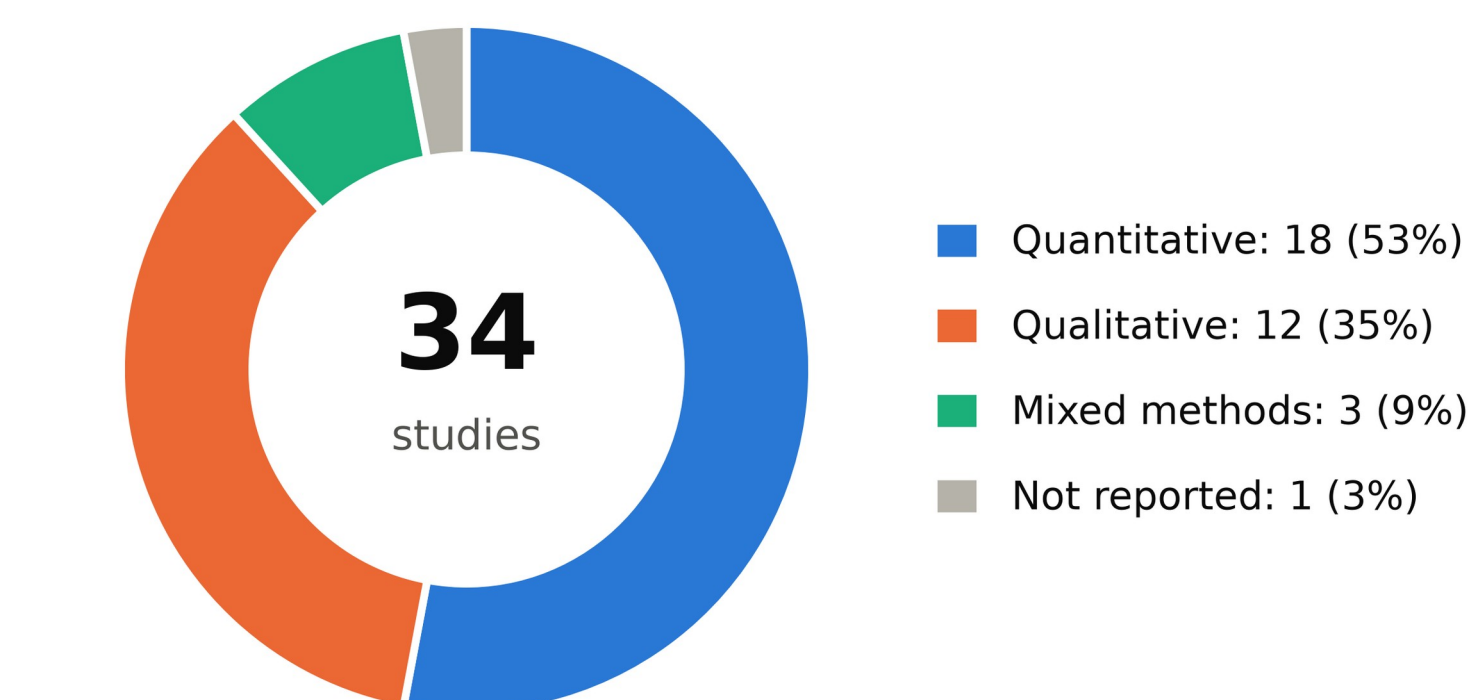


Fig 4: Study Design Distribution
Design of one study could not be determined from its abstract

Results, Discussion and Conclusion

Results

Of 516 records, 324 remained after deduplication and 34 studies were provisionally included: 18 quantitative, 12 qualitative, 3 mixed-methods, and 1 not reported. Psychosexual and relational challenges were most common (71%), followed by QoL, body image and self-esteem (53%), broader distress (53%), and depression and anxiety (29%). Healthcare gaps (9 studies) included providers' limited knowledge of MRKH, insensitive communication at diagnosis, and scarce information, psychological and fertility counseling.

Discussion

Findings on depression and anxiety were mixed: several studies reported elevated symptoms, while others found levels similar to controls. This aligns with earlier evidence^{5–7} and a prior systematic review highlighting sexual esteem and genital image.⁸ Coping relied largely on peer support and self-management rather than systematic clinical pathways.

Conclusion

MRKH-related psychosocial burden is substantial yet under-integrated into care models. Multidisciplinary, mental health-inclusive care and greater research investment are urgently needed, particularly in underrepresented and low-resource settings.

Limitations

- Provisional: AI-assisted screening and abstract-level extraction pending full-text verification
- Review limited to 2019–2026
- Heterogeneous designs and measures limit comparison
- Few studies from Africa; none from South America

Public Health Implications

- Integrate mental health screening into MRKH diagnostic pathways
- Develop multidisciplinary care protocols including psychology and social work
- Expand research investment in LMIC and underrepresented settings
- Build peer support infrastructure within reproductive health services
- Train providers in psychosocially informed MRKH care

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